

A simple model of the basic economic problem: the production possibilities frontier and opportunity costs

**ECON 101: Introduction to
Microeconomics**

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Production Possibilities and Opportunity Cost

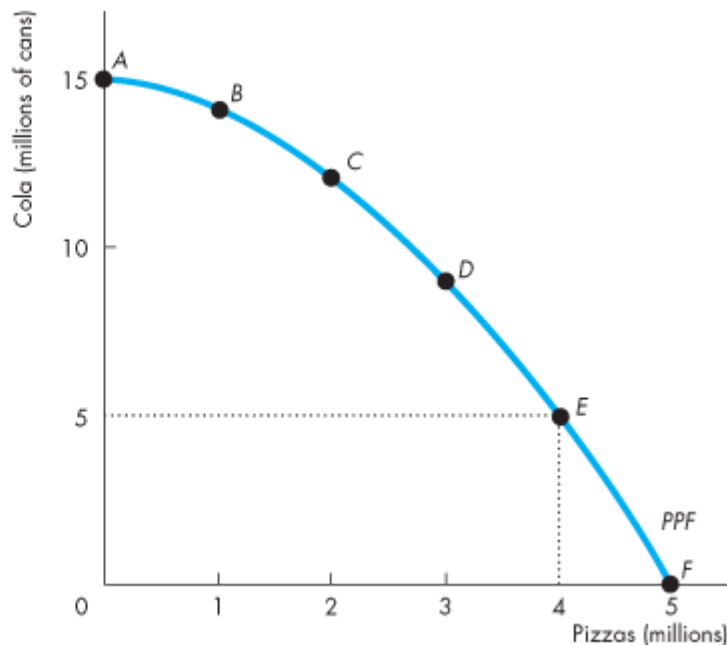
- The **production possibilities frontier** (*PPF*) is the boundary between those combinations of goods and services that can be produced and those that cannot.
- To illustrate the *PPF*, we focus on two goods at a time and hold the quantities of all other goods and services constant.
- That is, we look at a model economy in which everything remains the same (*ceteris paribus*) except the two goods we're considering.



Production Possibilities and Opportunity Cost

Production Possibilities Frontier

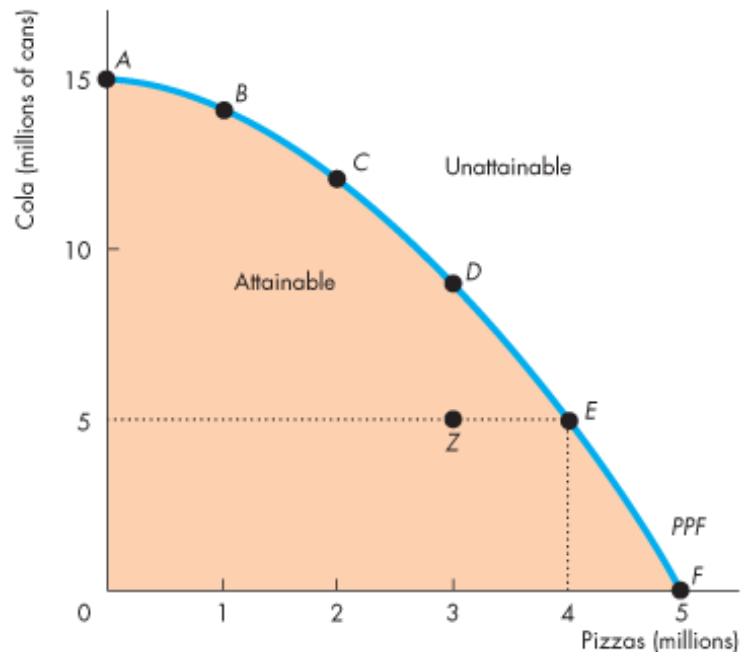
Possibility	Pizzas (millions)		Cola (millions of cans)
A	0	and	15
B	1	and	14
C	2	and	12
D	3	and	9
E	4	and	5
F	5	and	0



Production Possibilities and Opportunity Cost

Production Possibilities Set and Production Possibilities Frontier

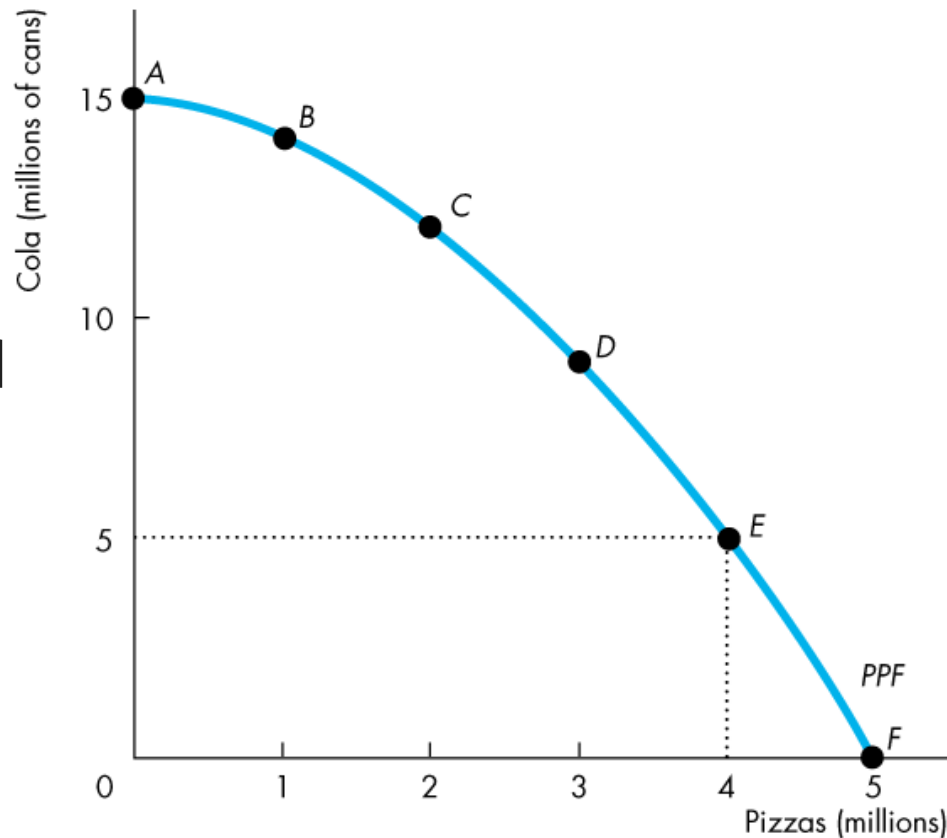
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Production Possibilities and Opportunity Cost

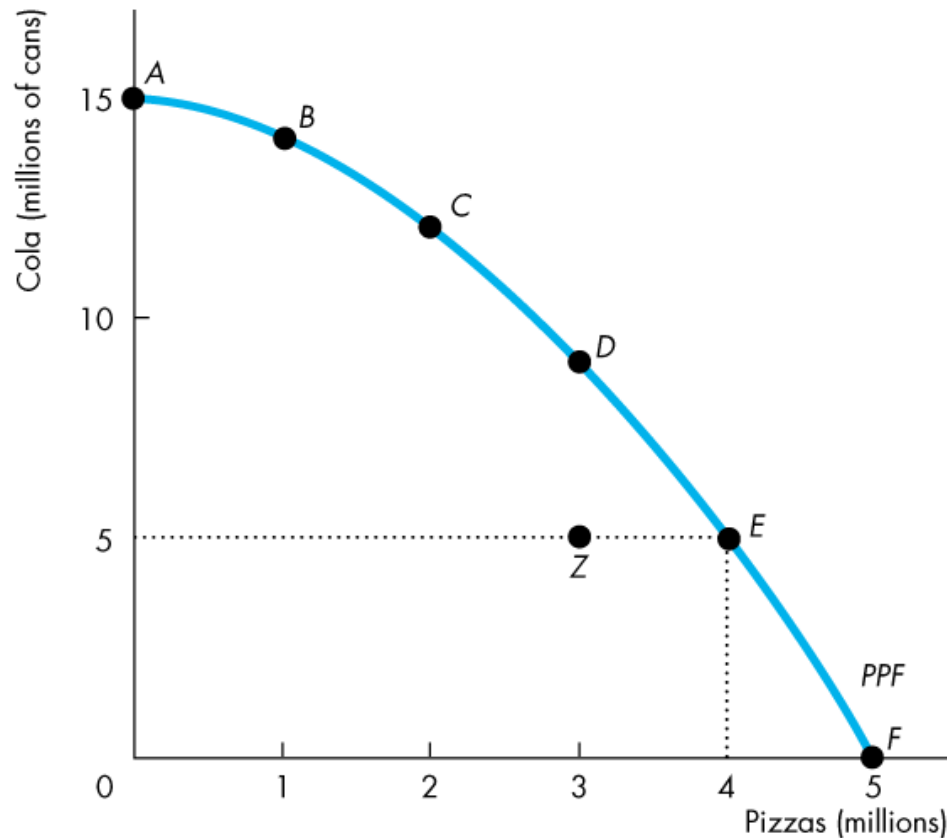
Production Efficiency

- We achieve production efficiency if we cannot produce more of one good without producing less of some other good.
- Points on the frontier are efficient.



Production Possibilities and Opportunity Cost

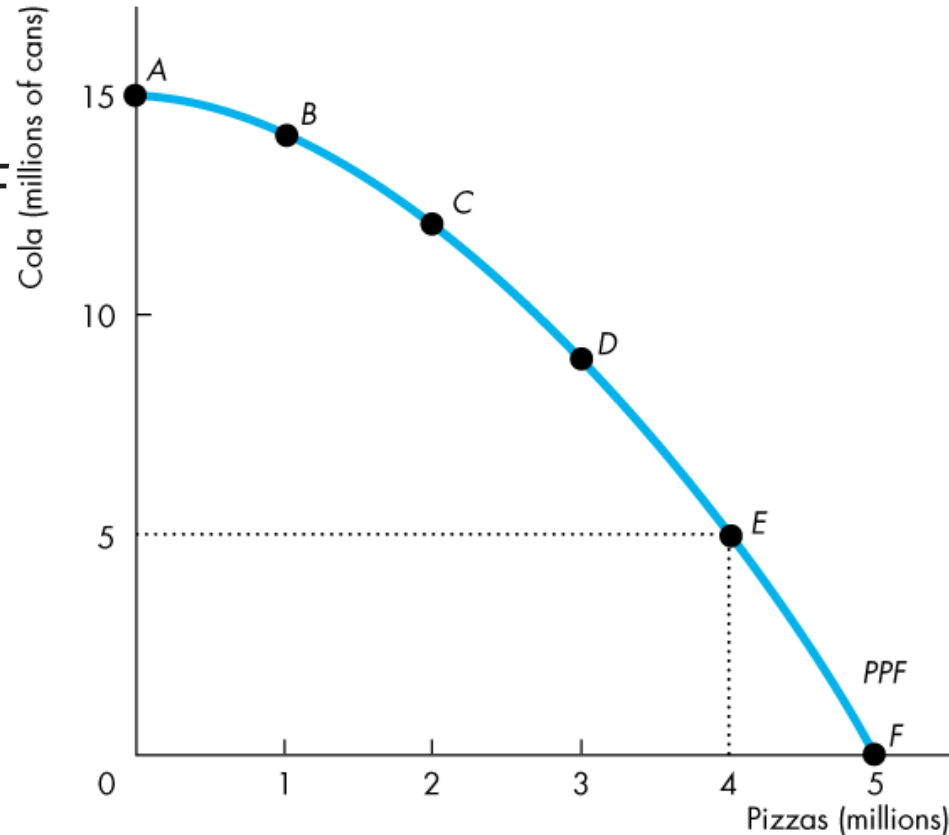
- Any point inside the frontier, such as Z, is inefficient.
- At such a point, it is possible to produce more of one good without producing less of the other good.
- At Z, resources are either unemployed or misallocated.



Production Possibilities and Opportunity Cost

Tradeoff Along the PPF

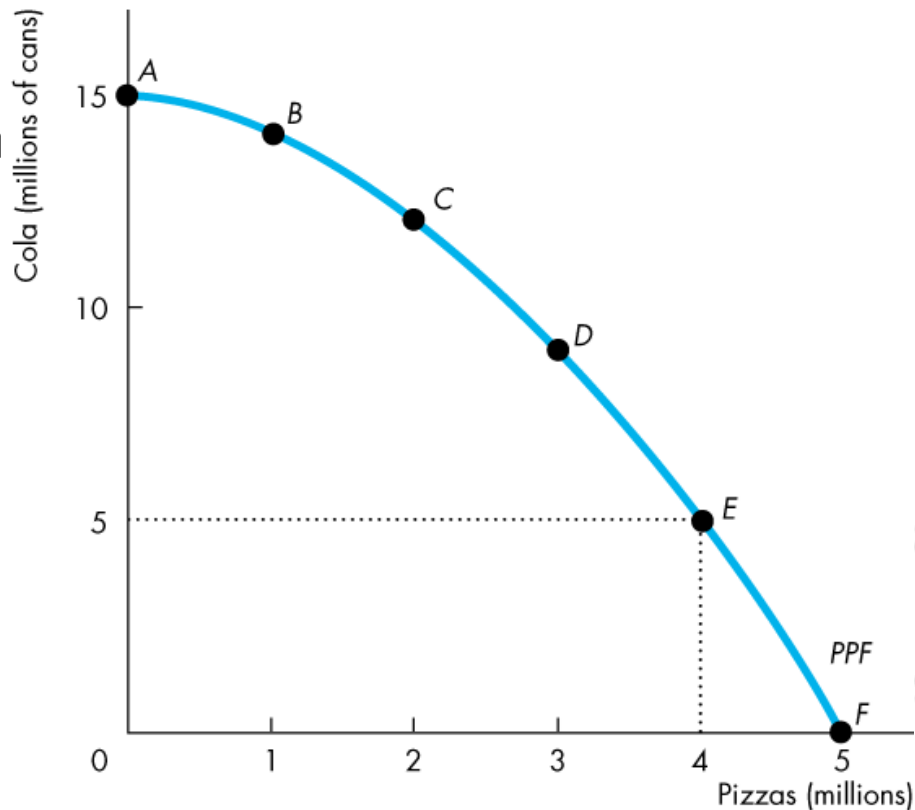
- Every choice along the PPF involves a tradeoff.
- On this PPF, we must give up some cola to get more pizzas or give up some pizzas to get more cola.



Production Possibilities and Opportunity Cost

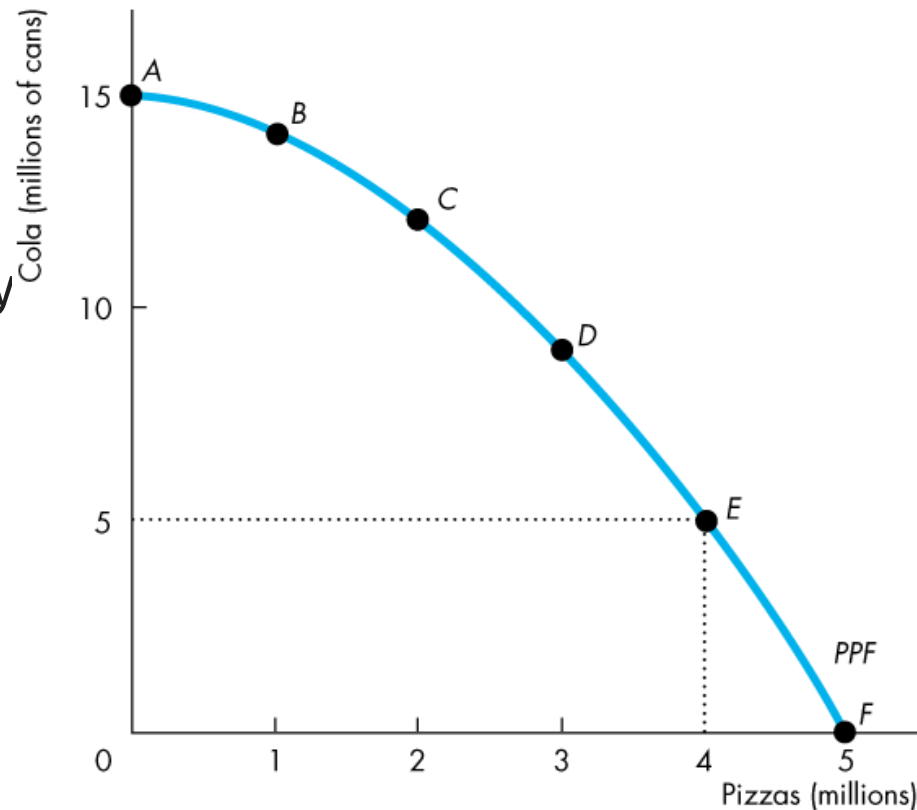
Opportunity Cost:

- The opportunity cost of an action is the highest-valued alternative forgone.
- As we move down along the PPF,
- we produce more pizzas, but the quantity of cola we can produce decreases.
- The opportunity cost of a pizza is the cola forgone.



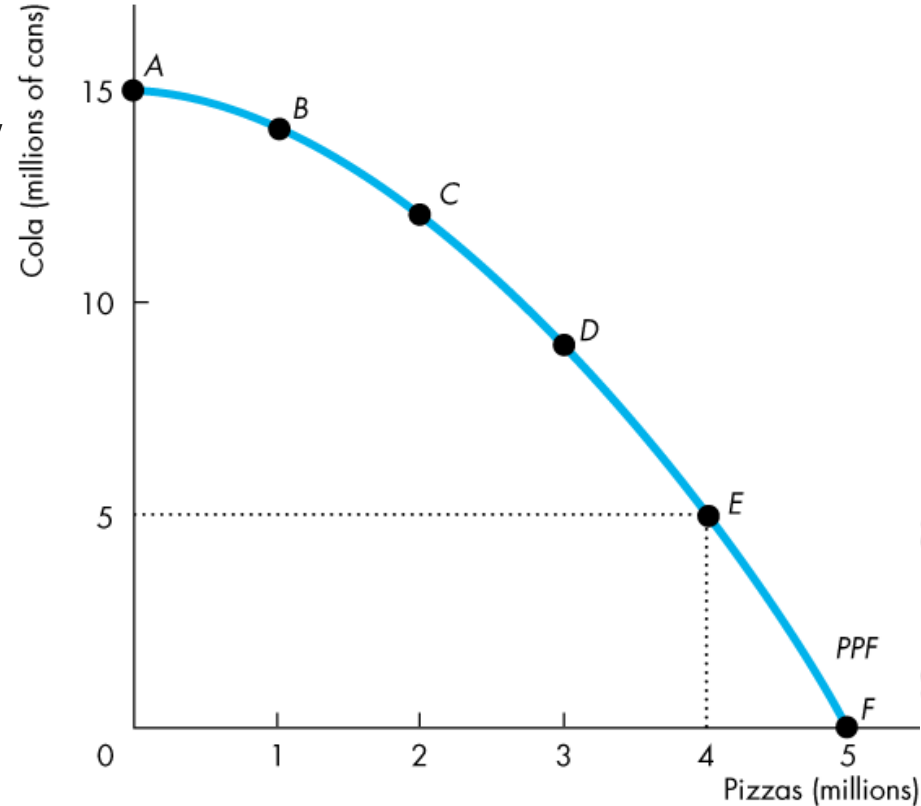
Production Possibilities and Opportunity Cost

- In moving from E to F:
- The quantity of pizzas increases by 1 million.
- The quantity of cola decreases by 5 million cans.
- The opportunity cost of the fifth 1 million pizzas is 5 million cans of cola.
- One of these pizzas costs 5 cans of cola.



Production Possibilities and Opportunity Cost

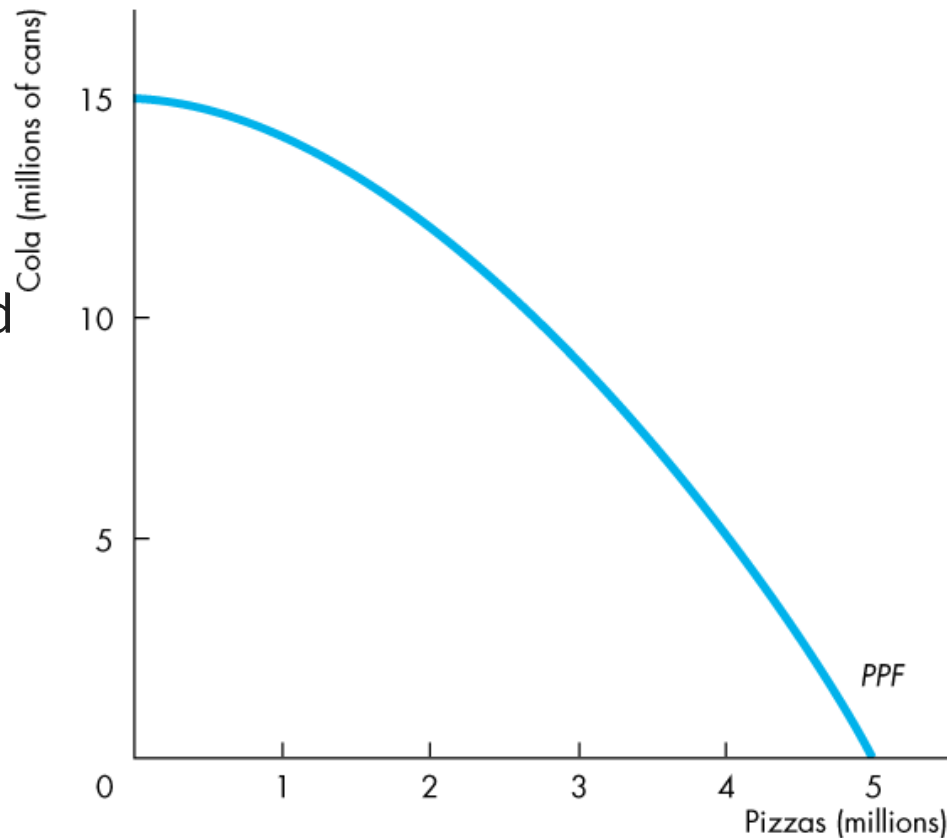
- In moving from F to E:
- The quantity of cola increases by 5 million cans.
- The quantity of pizzas decreases by 1 million.
- The opportunity cost of the first 5 million cans of cola is 1 million pizzas.
- One of these cans of cola costs $\frac{1}{5}$ of a pizza.



Production Possibilities and Opportunity Cost

Increasing Opportunity Cost

- Because resources are not equally productive in all activities, the PPF bows outward
- The outward bow of the PPF means that as the quantity produced of each good increases, so does its opportunity cost.

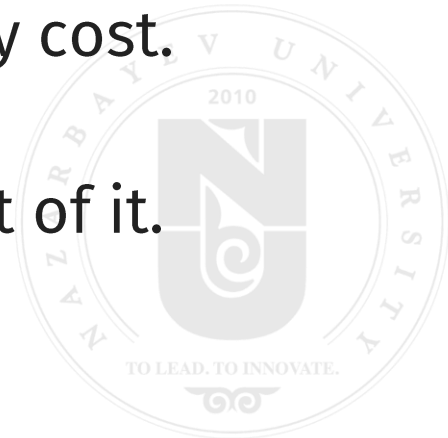


Using Resources Efficiently

- All the points along the PPF are efficient.
- To determine which of the alternative efficient quantities to produce, we compare costs and benefits.

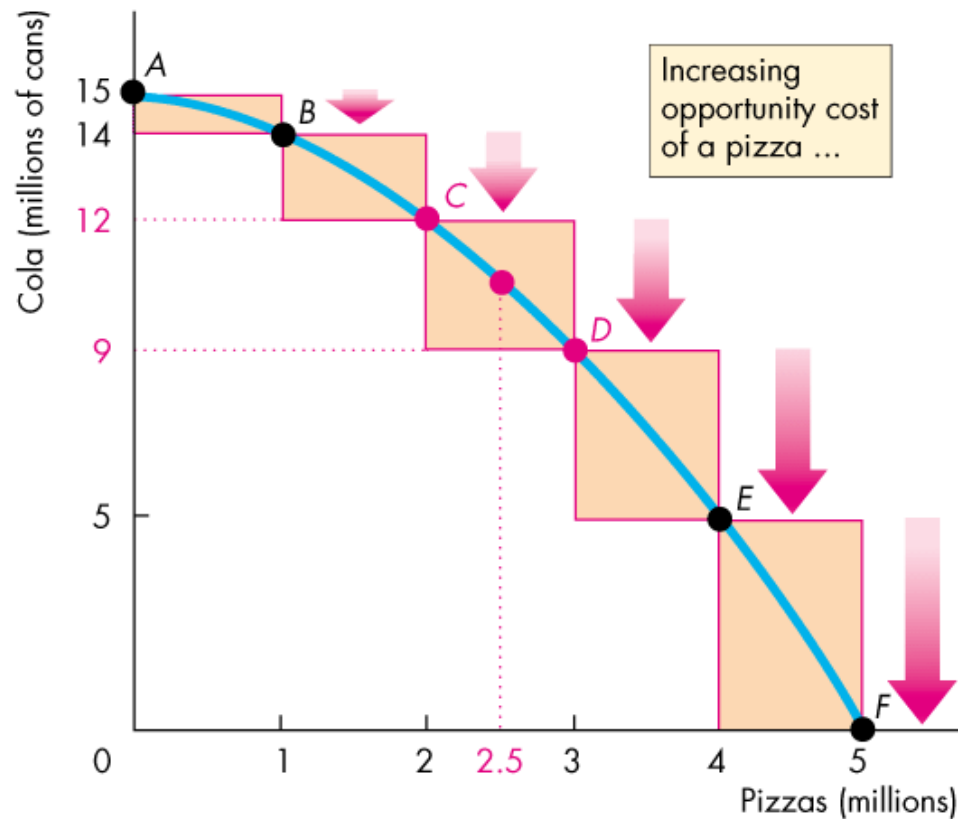
The PPF and Marginal Cost

- The **slope** of the PPF determines opportunity cost.
- The marginal cost of a good or service is the opportunity cost of producing one more unit of it.



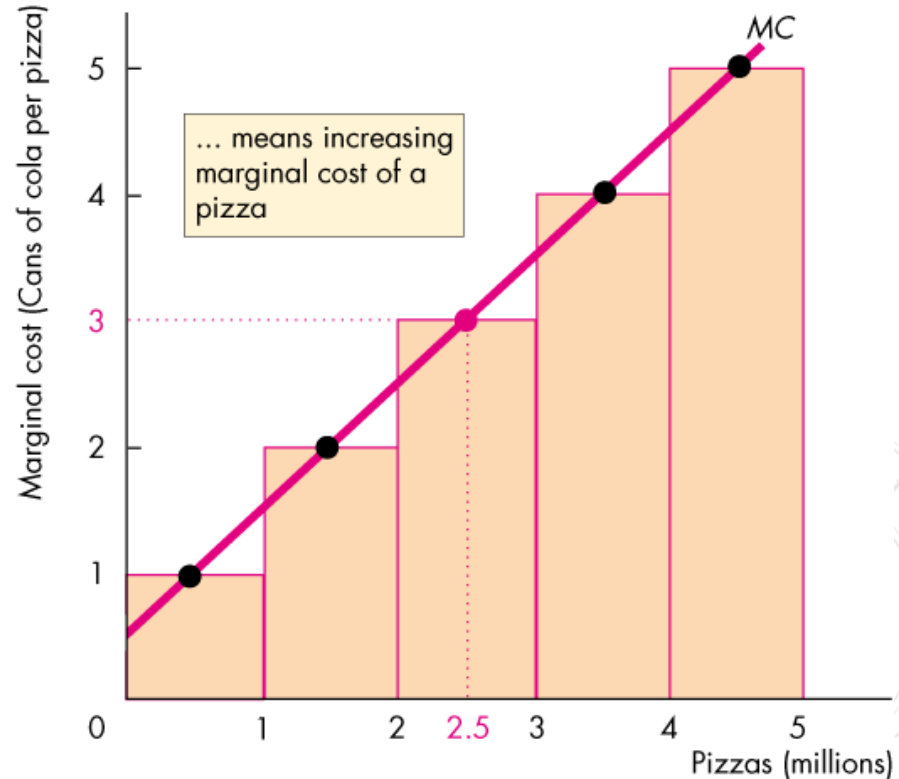
Using Resources Efficiently

- The figure illustrates the marginal cost of a pizza.
- As we move along the PPF, the opportunity cost of a pizza increases.
- The opportunity cost of producing one more pizza is the marginal cost of a pizza.



Using Resources Efficiently

- The bars illustrate the increasing opportunity cost of a pizza.
- The black dots and the line MC show the marginal cost of producing a pizza.
- The MC curve passes through the middle point of each bar.



(b) Marginal cost

Using Resources Efficiently

Preferences and Marginal Benefit

- Preferences are a description of a person's likes and dislikes.
- To describe preferences, economists use the concepts of marginal benefit and the marginal benefit curve.
- The marginal benefit of a good or service is the benefit received from consuming one more unit of it.
- We measure marginal benefit by the amount that a person is willing to pay for an additional unit of a good or service.



Using Resources Efficiently

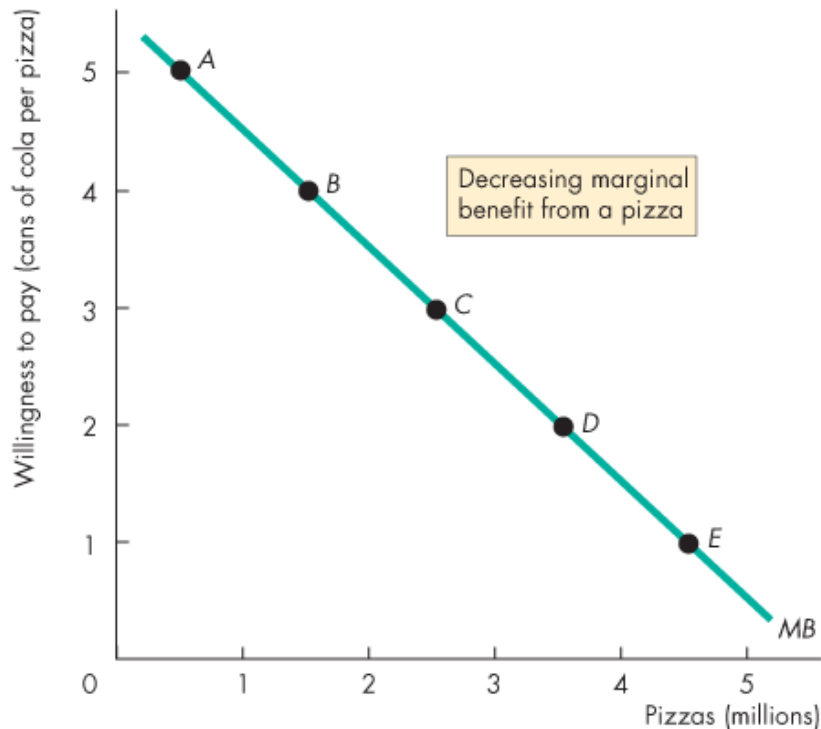
- It is a general principle that:
- The more we have of any good, the smaller is its marginal benefit and ...
- the less we are willing to pay for an additional unit of it.
- We call this general principle the *principle of decreasing marginal benefit*.
- The **marginal benefit curve** shows the relationship between the marginal benefit of a good and the quantity of that good consumed.



Using Resources Efficiently

- The line through the points shows the marginal benefit from a pizza.

Possibility	Pizzas (millions)	Willingness to pay (cans of cola per pizza)
A	0.5	5
B	1.5	4
C	2.5	3
D	3.5	2
E	4.5	1



Using Resources Efficiently

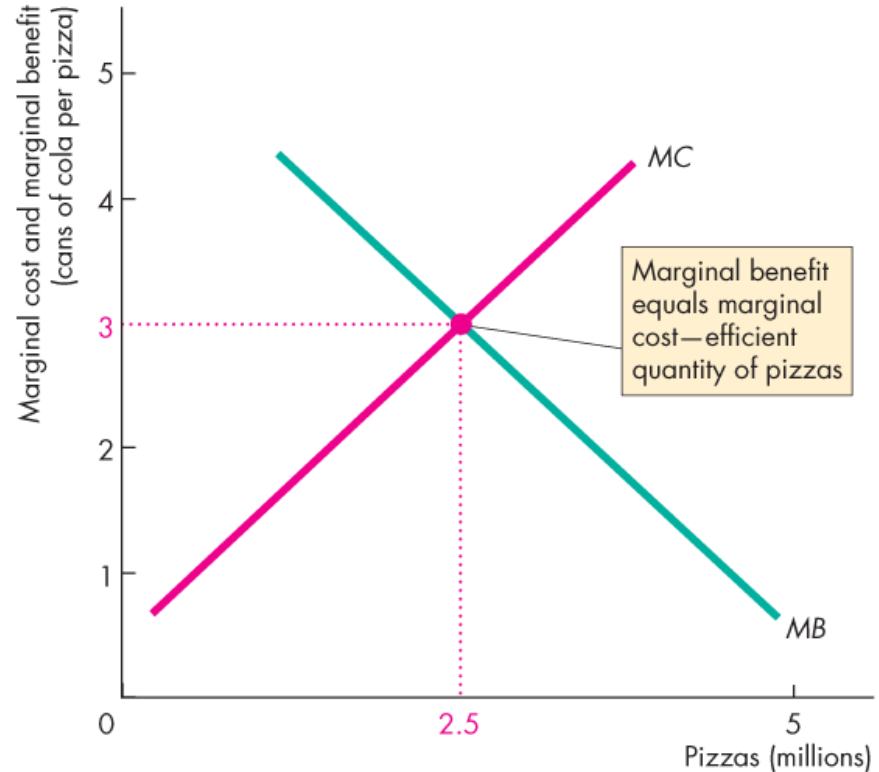
Allocative Efficiency

- When we cannot produce more of any one good without giving up some other good, we have achieved production efficiency.
- We are producing at a point on the PPF.
- When we cannot produce more of any one good without giving up some other good that we value more highly, we have achieved allocative efficiency.
- We are producing at the point on the PPF that we prefer above all other points.

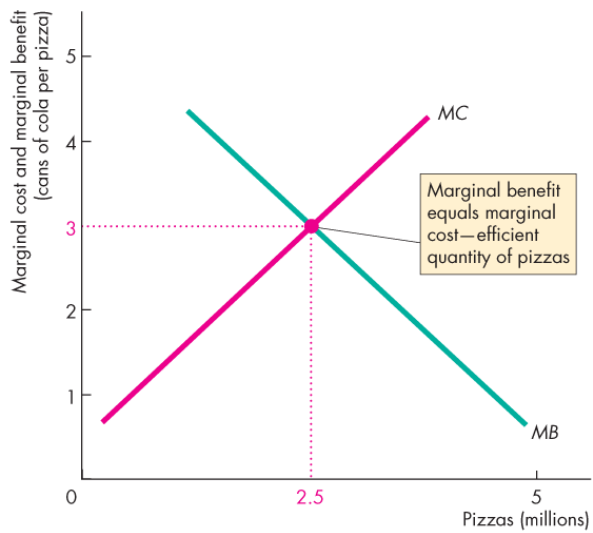


Using Resources Efficiently

- The point of allocative efficiency is the point on the PPF at which marginal benefit equals marginal cost.
- This point is determined by the quantity at which the marginal benefit curve intersects the marginal cost curve.

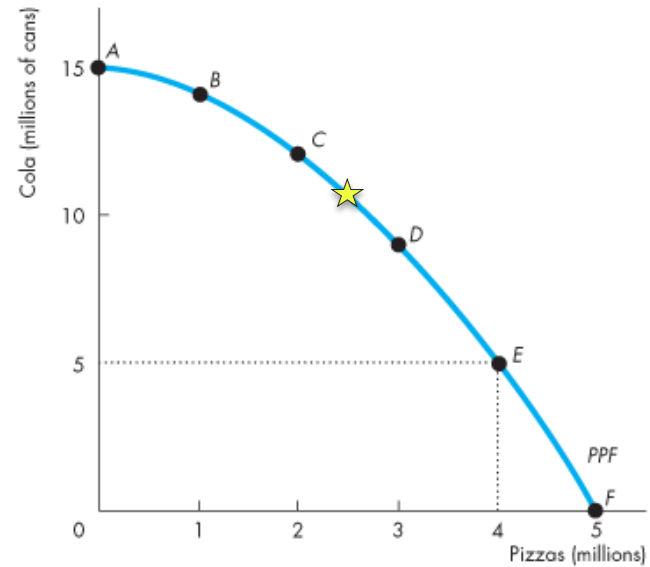


(b) Marginal benefit equals marginal cost



(b) Marginal benefit equals marginal cost

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Economic Growth

- The expansion of production possibilities—an increase in the standard of living—is called economic growth.
- Two key factors influence economic growth:
 - Technological change
 - Capital accumulation
- Technological change is the development of new goods and of better ways of producing goods and services.
- Capital accumulation is the growth of capital resources, which includes human capital.



Economic Growth

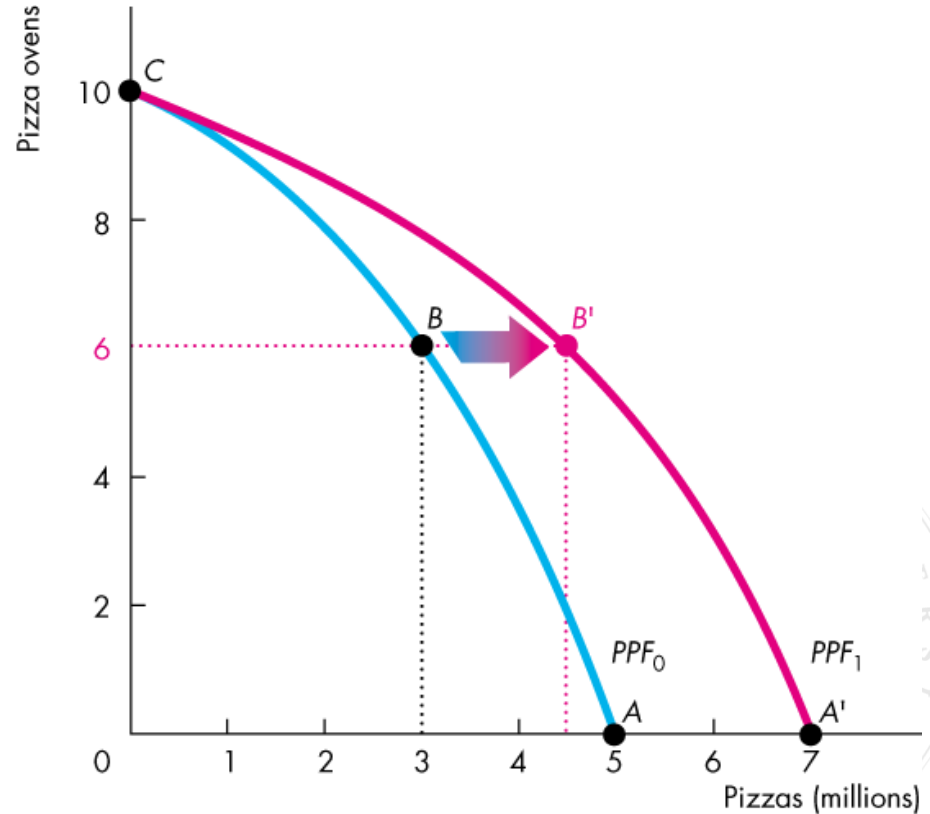
The Cost of Economic Growth

- To use resources in research and development and to produce new capital, we must decrease our production of consumption goods and services.
- So economic growth is not free.
- The opportunity cost of economic growth is less current consumption.



Economic Growth

- We can produce pizzas or pizza ovens along PPF_0 .
- By using some resources to produce pizza ovens today, the PPF shifts outward in the future.



Gains from Trade

Comparative Advantage and Absolute Advantage

- A person has a comparative advantage in an activity if that person can perform the activity at a lower opportunity cost than anyone else.
- A person has an absolute advantage if that person is more productive than others.
- Absolute advantage involves comparing productivities while comparative advantage involves comparing opportunity costs.



Gains from Trade

Joe's Smoothie Bar

In an hour, Joe can produce 6 smoothies or 30 salads.

Joe's opportunity cost of producing 1 smoothie is 5 salads.

Joe's opportunity cost of producing 1 salad is $\frac{1}{5}$ smoothie.

Joe spends 10 minutes making salads and 50 minutes making smoothies, so he produces 5 smoothies and 5 salads an hour.

TABLE 2.1 Joe's Production Possibilities

Item	Minutes to produce 1	Quantity per hour
Smoothies	10	6
Salads	2	30



Gains from Trade

Liz's Smoothie Bar

In an hour, Liz can produce 30 smoothies or 30 salads.

Liz's opportunity cost of producing 1 smoothie is 1 salad.

Liz's opportunity cost of producing 1 salad is 1 smoothie.

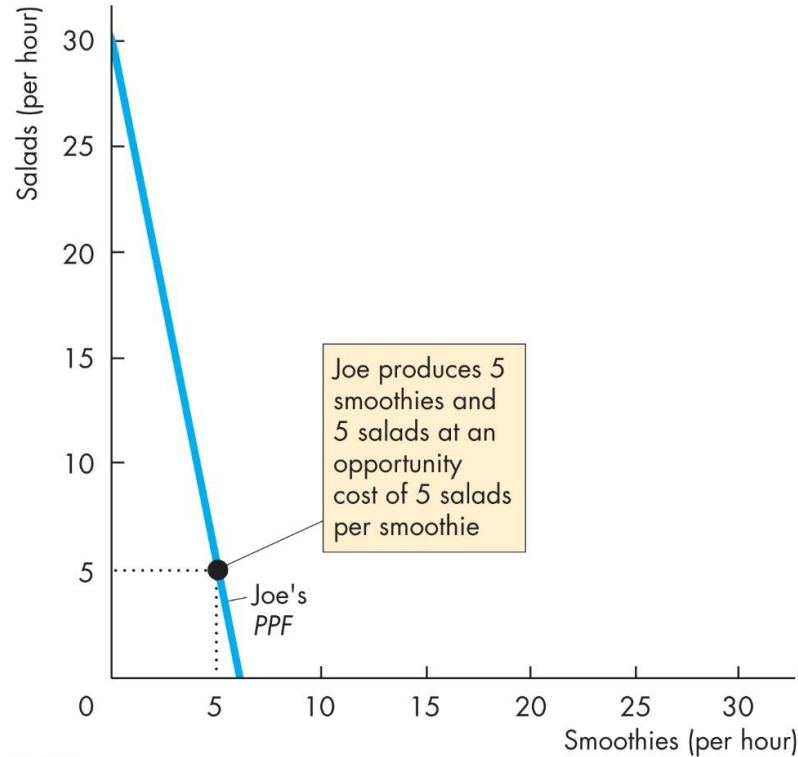
Liz's customers buy salads and smoothies in equal number, so she produces 15 smoothies and 15 salads an hour.

TABLE 2.2 Liz's Production Possibilities

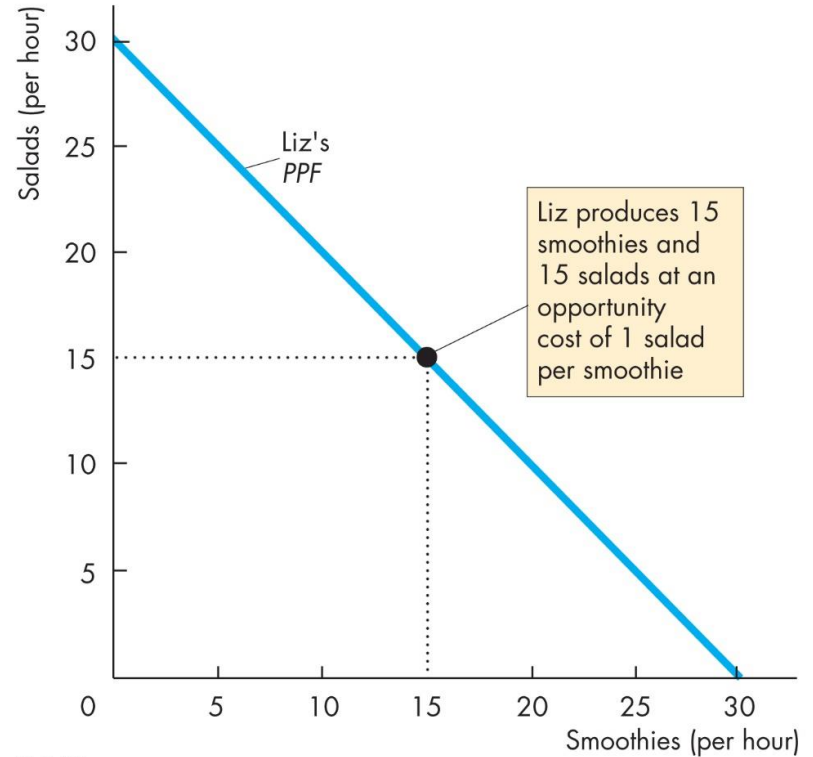
Item	Minutes to produce 1	Quantity per hour
Smoothies	2	30
Salads	2	30



Gains from Trade



(a) Joe



(b) Liz

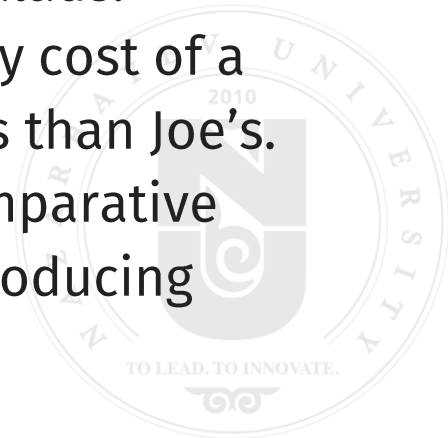
Gains from Trade

Joe's Comparative advantage

- Joe's opportunity cost of a salad is $\frac{1}{5}$ smoothie.
- Liz's opportunity cost of a salad is 1 smoothie.
- Joe's opportunity cost of a salad is less than Liz's.
- So Joe has a comparative advantage in producing salads.

Liz's Comparative advantage

- Liz's opportunity cost of a smoothie is 1 salad.
- Joe's opportunity cost of a smoothie is 5 salads.
- Liz's opportunity cost of a smoothie is less than Joe's.
- So Liz has a comparative advantage in producing smoothies.



Gains from Trade

Achieving the Gains from Trade

Liz and Joe produce the good in which they have a comparative advantage:

- Liz produces 30 smoothies and 0 salads.
- Joe produces 30 salads and 0 smoothies.

Liz and Joe trade:

- Liz sells Joe 10 smoothies and buys 20 salads.
- Joe sells Liz 20 salads and buys 10 smoothies.

After trade:

- Liz has 20 smoothies and 20 salads.
- Joe has 10 smoothies and 10 salads.

Gains from trade:

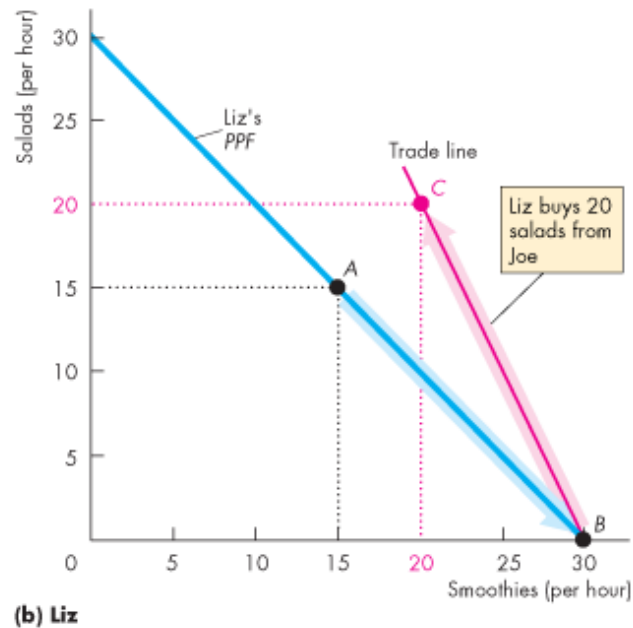
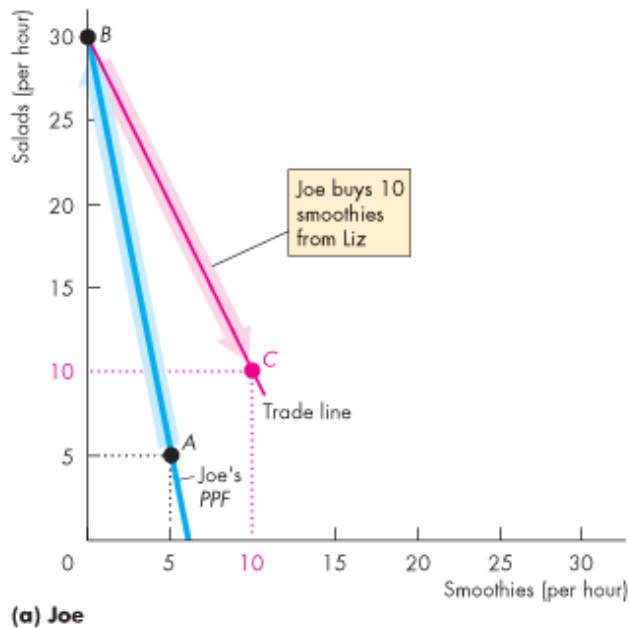
- Liz gains 5 smoothies and 5 salads an hour
- Joe gains 5 smoothies and 5 salads an hour

TABLE 2.3 Liz and Joe Gain from Trade

(a) Before trade	Liz	Joe
Smoothies	15	5
Salads	15	5
(b) Specialization	Liz	Joe
Smoothies	30	0
Salads	0	30
(c) Trade	Liz	Joe
Smoothies	sell 10	buy 10
Salads	buy 20	sell 20
(d) After trade	Liz	Joe
Smoothies	20	10
Salads	20	10
(e) Gains from trade	Liz	Joe
Smoothies	+5	+5
Salads	+5	+5

Gains from Trade

- Joe buys smoothies from Liz and moves to point C—a point outside his PPF.
- Liz buys salads from Joe and moves to point C—a point outside her PPF.





End of this lecture.
Read Chapter 2 from Parkin.